Predicting VA Cash-Out Refinance Share Across States Using Machine Learning: A Literature Review and Empirical Study

Abstract:

This report studies whether supervised machine learning (ML) models can forecast and explain state-level variation in the share of Veterans Affairs (VA)-backed loans that are classified as Cash-Out Refinance. The report has three goals: (i) motivate the importance of cash-out behavior through a structured review of the household-finance and mortgage literature; (ii) synthesize major research lines, from early work on optimal refinancing rules and borrower mistakes to modern studies on equity extraction and debt substitution; and (iii) evaluate a baseline ML forecasting pipeline on a small state-year panel (2020–2024) under strict out-of-time testing. Empirically, a persistence baseline is benchmarked against regularized linear regression and tree-based ensemble models. Initial results show that the lag-based persistence baseline is difficult to outperform using internal loan-composition features alone, motivating richer temporal features, careful tuning, and integration of external macro-housing drivers (rates, prices, labor markets) and explainability outputs in the final model.

Index Terms:

mortgage refinancing, cash-out refinance, home equity extraction, VA loans, state-level forecasting, machine learning, random forests, gradient boosting, SHAP.

I. DESCRIPTION OF TOPIC

Cash-out refinancing replaces an existing mortgage with a new mortgage and allows the borrower to withdraw a portion of accumulated home equity as cash. At the household level, cash-out proceeds may be used for debt consolidation, consumption smoothing, home improvements, or other liquidity needs. At the aggregate level, cash-out activity is connected to the transmission of interest-rate changes and to household balance sheet dynamics. This project focuses on forecasting and explaining variation in the cash-out share of VA lending across U.S. states and years.

II. MOTIVATION AND SCOPE

Forecasting cash-out share has operational value because it reflects shifts in product mix and borrower behavior that affect lender capacity planning and market strategy. From a research standpoint, cash-out refinancing is a high-signal behavior: it is sensitive to incentives (rates, home prices) but also mediated by frictions (fixed costs, inattention) and by constraints that affect whether equity can be extracted or substituted with other credit products. The VA program provides a practical, program-relevant context with standardized loan-type categories and public reporting [16].

Scope and out-of-scope: This report reviews research on refinancing optimality, inattention, equity extraction, and debt substitution. Borrower-level underwriting, fairness analyses,

and default prediction are out of scope due to the aggregate nature of the dataset. Causal estimation of policy shocks is also out of scope; the objective is forecasting and descriptive interpretation under realistic out-of-time validation.

III. FORMAL PROBLEM STATEMENT

Let s denote a state/territory and t denote year. The target is cash-out share y_{s,t} = CashOutCount_{s,t} / TotalCount_{s,t}. Given predictors X_{s,t-1} constructed from lagged features and state identifiers, the task is supervised regression: learn f(.) such that yhat_{s,t} = f(X_{s,t-1}). Evaluation is out-of-time: train on earlier years and test on a later year. Research questions: (1) which features are most predictive of cash-out share at the state-year level, and (2) do ML models meaningfully outperform strong persistence baselines?

IV. LITERATURE REVIEW

A. Optimal refinancing rules and option value

The theoretical foundations of refinancing analysis trace to the mortgage prepayment literature of the early 1980s. Dunn and McConnell first modeled the mortgage as a callable bond, deriving the borrower's optimal prepayment rule under stochastic interest rates as a free-boundary problem [19]. Schwartz and Torous extended this framework by introducing a proportional hazard model that incorporated both rational option exercise and empirically observed prepayment burnout, providing the first econometric bridge between structural theory and observed behavior [20]. These early models established the core tension in the literature: observed prepayment deviates systematically from the frictionless optimum, motivating decades of subsequent work on costs, inattention, and constraints. Building on this tradition, Agarwal, Driscoll, and Laibson show that widely used net-present-value rules can ignore option value and diverge materially from optimal decisions [10]. Agarwal et al. derive a closed-form refinancing rule, providing a benchmark for how refinancing should respond to rate gaps under an optimal policy [11]. Importantly, the closed-form solution in [11] can be seen as a tractable approximation to the full dynamic programming problem posed by Dunn and McConnell [19]: where the early work required numerical methods, [11] yields an explicit threshold in terms of rate gap, fixed costs, and volatility. These models clarify why refinancing thresholds can be non-linear in rates and costs, and they motivate modeling that allows for regime shifts.

B. Borrower mistakes, learning, and inattention

Empirical evidence shows systematic deviations from optimal refinancing. Agarwal, Rosen, and Yao document that a large share of borrowers refinance sub-optimally and classify errors as commission and omission, with financially sophisticated borrowers making smaller mistakes [1]. Their empirical approach is reduced-form: they measure deviations from the Agarwal et al. threshold rule [11] and correlate them with borrower characteristics. Berger, Milbradt, Tourre, and Vavra provide a structural complement: they model sporadic attention explicitly and derive statistics to quantify inattention frictions, explaining why borrowers may fail to refinance even when profitable [11]. Crucially, the Berger et al. model implies that the magnitude of inattention-driven delay is increasing in the frequency of profitable refinancing opportunities, which has direct implications for forecasting: in environments

where rate incentives are large and persistent (as in 2020–2021), attention models predict catch-up behavior that would eventually be captured in lagged features. Together, these works suggest that inertia is not noise; it is a structural feature that should be reflected in forecasting baselines and features.

Connection to this project: At the state-year level, persistence can reflect slow-moving borrower composition, lender presence, and local constraints, but it can also reflect inattention frictions that dampen responsiveness to rates. This justifies benchmarking against y_{t-1} and evaluating whether additional predictors yield robust lift.

C. Equity extraction, house prices, and leverage

The equity extraction literature ties borrowing against housing wealth to aggregate leverage and consumption. Mian and Sufi link house price gains to increased home equity-based borrowing and the buildup of household leverage prior to the financial crisis [2]. Bhutta and Keys estimate strong responses of equity extraction to mortgage rate declines and show that house price growth amplifies extraction responses [3]. A forecasting implication is that models should include both rate and price dynamics, with an emphasis on changes (deltas) that can capture regime transitions.

Duca, Muellbauer, and Murphy find that financial literacy is associated with lower propensity to withdraw housing equity [12]. This supports a view that cross-state differences can persist even under common macro conditions, and it motivates state fixed effects or state embeddings in ML models.

D. Conditional cash-out decisions, frictions, and debt substitution

Hossain, Livshits, and Wardius analyze why cash-out refinancing remains low even among borrowers who refinance and hold high-interest debt, emphasizing frictions that inhibit equity extraction [4], [5]. Their explanation is friction-based: borrowers face information costs, transaction costs, or lender constraints that prevent extraction even when it would be financially beneficial. Anenberg, Scharlemann, and van Straelen offer a complementary but distinct mechanism: changes in cash-out refinancing may be offset by changes in other borrowing (substitution), weakening net borrowing responses [6], [7]. The two explanations have different forecasting implications. If Hossain et al. are correct, cash-out share should respond sluggishly to rate incentives because frictions dominate—consistent with a strong persistence baseline. If Anenberg et al. are correct, aggregate cash-out share may be partially predictable from credit market conditions even when household-level frictions are present, since substitution patterns shift with credit availability. Together, these studies indicate that cash-out share depends not only on refinancing incentives but also on the broader credit environment and substitution opportunities.

Connection to this project: In aggregate data, substitution is not directly observed; it appears as weaker-than-expected sensitivity of cash-out share to rates. Therefore, interpreting ML feature importance requires caution: low importance for rates does not imply rates do not matter; it may reflect substitution or constraints varying across states.

E. VA program context and measurement

VA reporting provides standardized categories and public summaries. VA “Loan Volume by State” reports depict purchase, IRRRL, and

cash-out refinance volumes and amounts by state and territory [16]. VA guidance also distinguishes cash-out refinance types in VA-to-VA refinances and outlines requirements such as seasoning and net tangible benefit tests [17]. These rules can create threshold effects (e.g., classification boundaries or seasoning constraints) that interact with market conditions, reinforcing the value of flexible ML models.

F. ML methods and best practices

The ML literature provides the methodological backbone for the experimental section. Random forests reduce variance via ensembling of randomized trees [13], while gradient boosting builds additive models via functional gradient descent and is known to capture complex interactions in structured data [14]. Regularization is important in small samples; the lasso introduced by Tibshirani demonstrates how penalization can stabilize estimates and support feature selection [18]. Schultz and Fabozzi demonstrate that ML models—particularly gradient boosting and neural networks—meaningfully outperform traditional prepayment models when rich macro and loan-level features are available [15], reinforcing the importance of integrating exogenous drivers in the current pipeline. Best practices applied here include strict train/validation/test separation, time-aware evaluation, and avoiding hyperparameter tuning on the test set.

V. DATASET AND FEATURES

The empirical dataset is a 2020–2024 state-year panel (54 state/territory codes). Each record includes loan counts by type (Purchase, IRRRL, Cash-Out), total counts, and loan amount summaries. The target variable is Pct_Cashout. Predictors in the current phase are lag-1 versions of internal variables to reduce leakage when predicting a share. Planned exogenous predictors include mortgage rates, house price indices, and unemployment, motivated by equity extraction and substitution research [3], [6].

Table II. Key dataset fields used for modeling and analysis.

Category	Fields (examples)	Purpose
Identifiers	State, Year	Panel indexing; controls for location and time
Loan counts	Total_Count, Loan Count Cash-Out, Loan Count IRRRL, Loan Count Purchase	Construct shares; lag features
Shares	Pct_Cashout, Pct_IRRRL, Pct_Purchase	Prediction target and lagged persistence
Amounts	Total_Amount; Avg Loan Amount (by type)	Scale and composition signals; lag features
Flags	High_Volume	Segment indicator for robustness checks

VI. METHODS

Baseline: persistence yhat_{s,t} = y_{s,t-1}. ML models: Ridge regression (regularized linear), decision tree regression, random forest

regression, and gradient boosting regression. Preprocessing: one-hot encoding for State, numeric standardization for linear models. Evaluation: out-of-time split (train 2021–2023, test 2024) with lag-1 predictors to reduce leakage.

VII. HYPERPARAMETER TUNING AND VALIDATION PLAN

To adhere to ML best practices, hyperparameter tuning is conducted on training data only. A time-aware approach is used: within the 2021–2023 training window, models are tuned using cross-validation folds that respect time ordering (e.g., train 2021–2022, validate 2023) rather than random shuffles. Example tuning ranges: ridge alpha in [1e-3, 1e3]; random forest (n_estimators 200–1000, max_depth 2–10); gradient boosting (learning_rate 0.01–0.2, n_estimators 200–2000, max_depth 2–5). The 2024 holdout year is used only once for final evaluation.

VIII. EXPERIMENTAL RESULTS

Metrics: MAE, RMSE, and R^2. MAE is average absolute error; RMSE penalizes large errors; R^2 measures improvement over predicting the mean. Models are trained on 2021–2023 and tested on 2024.

Table III. Out-of-time performance on the 2024 holdout year.

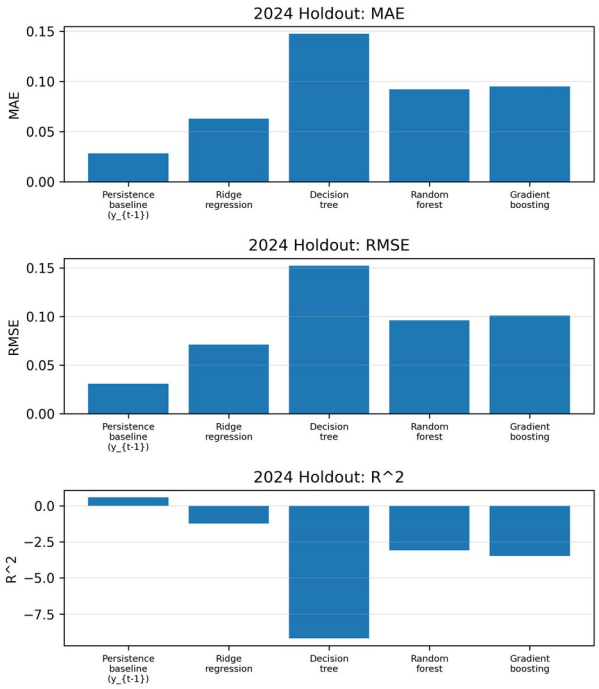


Fig. 1. Model comparison on 2024 holdout (MAE, RMSE, and R^2).

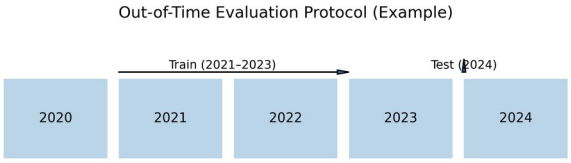


Fig. 2. Out-of-time evaluation protocol (train 2021-2023, test 2024).

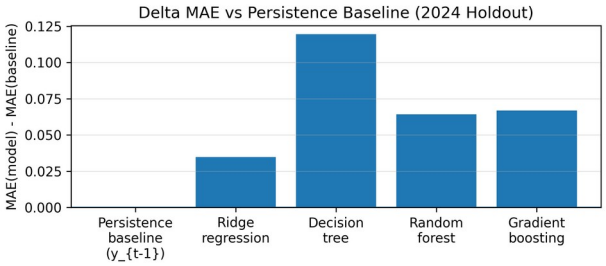


Fig. 3. Delta MAE relative to persistence baseline on 2024 holdout.

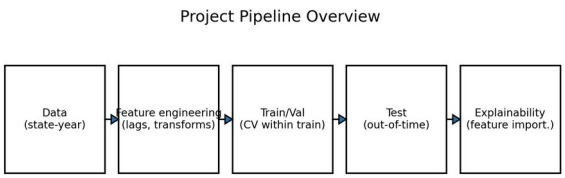


Fig. 4. Project pipeline overview from data to explainability outputs.

Model	MAE	RMSE	R^2
Persistence baseline (y_{t-1})	0.0281	0.0308	0.5818
Ridge regression	0.0629	0.0710	-1.2189
Decision tree	0.1475	0.1523	-9.1960
Random forest	0.0923	0.0963	-3.0803
Gradient boosting	0.0950	0.1010	-3.4836

Interpretation: The persistence baseline is the best performer. This indicates that state-level cash-out share is highly autocorrelated year-to-year in the current panel. Under small samples and limited exogenous signal, flexible tree ensembles can overfit and perform worse than a simple baseline. The result is actionable: to beat persistence, the model likely needs exogenous macro-housing drivers and stronger regularization.

IX. DISCUSSION AND ANALYSIS

The empirical results reinforce the literature's emphasis on persistence and frictions in refinancing. Inattention and fixed costs can reduce responsiveness to incentives, and state-level heterogeneity can be persistent across years. The strong lag baseline suggests that, at a one-year horizon and at a state aggregation level, structural differences dominate. However, the data contain a striking pattern that motivates the importance of macro drivers: mean cash-out share nearly doubled from approximately 12% in 2021 to 27% in 2022 before declining to 17% by 2024. This spike, visible in the state-year panel, aligns closely with the 2021–2022 interest rate cycle, during which borrowers with low-rate mortgages may have turned to cash-out refinancing as an alternative to rate-and-term refinancing when the rate incentive reversed. Several high-equity states (e.g., AZ, CA) exceeded 40% cash-out share in 2022. This pattern is precisely the regime transition that the equity extraction literature anticipates [3], and it suggests that a model incorporating lagged rate changes and house price growth could meaningfully improve on persistence during transition years, even if persistence dominates in stable years. The equity extraction and debt-substitution literatures suggest that rate and price dynamics matter especially during these regime shifts [3], [6]. Therefore, the most promising direction is to add macro-housing predictors and model both level and change effects.

Explainability will be used to connect predictive drivers to plausible mechanisms. In the final model, global feature importance will identify which variables drive cash-out share nationally, while local explanations will be used to understand states with unusually high errors or unusual trends. These explanations will be interpreted in light of the literature: e.g., rate sensitivity consistent with equity extraction studies, or muted sensitivity consistent with substitution effects.

X. FUTURE WORK

The final phase will integrate external macro predictors (mortgage rates, FHFA house price indices, unemployment) and evaluate boosted learners with early stopping. Evaluation will use rolling-origin testing across multiple years (e.g., train through 2022, test 2023; train through 2023, test 2024) to measure stability. Additional outputs include calibration checks and state-level error profiling.

XI. LIMITATIONS AND ETHICS

Limitations include small sample size, aggregation bias, and potential measurement differences across reporting periods. The analysis is forecasting-oriented and does not identify causal effects. The work uses aggregate data and no personally identifiable information; nevertheless, interpretation should avoid extrapolating state averages to individual borrowers.

XII. CONCLUSION

This report synthesizes key research on refinancing, equity extraction, and debt substitution, and evaluates a baseline ML forecasting pipeline for VA cash-out share using a state-year panel. Initial experiments show that a lag-based persistence baseline is difficult to beat with internal features alone. The final submission will incorporate macro-housing predictors, tuned boosted models, figures, and explainability artifacts to deliver a defensible assessment of predictability and drivers.

APPENDIX A. REPRODUCIBILITY AND SUBMISSION ARTIFACTS

The final submission includes: (i) this IEEE-format report; and (ii) code files converted from Jupyter to Python scripts with meaningful names. Figures are exported as PNG files and inserted into the report with numbered captions.

REFERENCES

[1] S. Agarwal, R. J. Rosen, and V. W. Yao, “Why do borrowers make mortgage refinancing mistakes?” FRB Chicago Working Paper, 2013. Available: https://www.chicagofed.org/-/media/publications/working-papers/2013/wp2013-02-pdf.pdf

[2] A. Mian and A. Sufi, “House prices, home equity-based borrowing, and the U.S. household leverage crisis,” American Economic Review, vol. 101, no. 5, pp. 2132–2156, 2011. Available: https://www.aeaweb.org/articles?id=10.1257/aer.101.5.2132

[3] N. Bhutta and B. J. Keys, “Interest rates and equity extraction during the housing boom,” American Economic Review, vol. 106, no. 7, pp. 1742–1774, 2016. Available: https://www.aeaweb.org/articles?id=10.1257/aer.20140040

[4] M. Hossain, I. Livshits, and C. Wardius, “Not cashing in on cashing out: An analysis of low cash-out refinance rates,” FRB Philadelphia Working Paper 23-04, 2023. Available: https://www.philadelphiafed.org/-/media/frbp/assets/working-papers/2023/wp23-04.pdf

[5] M. Hossain, I. Livshits, and C. Wardius, “An analysis of low cash-out refinance rates,” FRB Philadelphia Working Paper 26-01, 2026. Available: https://www.philadelphiafed.org/-/media/FRBP/Assets/working-papers/2026/wp26-01.pdf

[6] E. Anenberg, T. Scharlemann, and E. van Straelen, “Debt substitution and the cash-out refinance channel of monetary policy,” FEDS Working Paper, 2023. Available:

https://www.federalreserve.gov/econres/feds/files/2023073pap.pdf

[7] E. Anenberg, T. Scharlemann, and E. van Straelen, “Debt substitution and the cash-out refinance channel of monetary policy,” American Economic Review, 2025. Available: https://www.aeaweb.org/articles?id=10.1257/aer.20231264

[8] P. Purohit, “Explainable machine learning for fallout prediction in the mortgage industry,” Journal of Risk and Financial Management, 2024. Available: https://www.mdpi.com/1911-8074/17/10/431

[9] S. Agarwal, J. Driscoll, and D. Laibson, “A closed-form solution for optimal mortgage refinancing,” NBER Working Paper 13487, 2007. Available: https://www.nber.org/system/files/working_papers/w13487/w13487.pdf

[10] S. Agarwal et al., “Optimal mortgage refinancing: A closed-form solution,” Journal of Money, Credit and Banking, 2013. Available: https://onlinelibrary.wiley.com/doi/10.1111/jmcb.12017

[11] D. W. Berger, K. Milbradt, F. Tourre, and J. S. Vavra, “Optimal mortgage refinancing with inattention,” NBER Working Paper 32447, 2024. Available: https://www.nber.org/papers/w32447

[12] J. V. Duca, J. Muellbauer, and A. Murphy, “Financial literacy and mortgage equity withdrawals,” Journal of Financial Stability, 2014. Available: https://www.sciencedirect.com/science/article/abs/pii/S0094119013000600

[13] L. Breiman, “Random forests,” Machine Learning, 2001. Available: https://www.stat.berkeley.edu/~breiman/randomforest2001.pdf

[14] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” Annals of Statistics, 2001. Available: https://projecteuclid.org/journals/annals-of-statistics/volume-29/issue-5/Greedy-function-approximation-A-gradient-boosting-machine/10.1214/aos/1013203451.full

[15] G. M. Schultz and F. J. Fabozzi, “Rise of the machines: Application of machine learning to mortgage prepayment modeling,” Journal of Fixed Income, 2022. Available: https://www.pm-research.com/content/iijfixinc/31/3/6

[16] U.S. Department of Veterans Affairs, “Loan Volume by State,” VA Home Loans. Available: https://www.benefits.va.gov/HOMELOANS/lender_state_volume.asp

[17] U.S. Department of Veterans Affairs, “Cash-Out Refinance User Guide,” 2018. Available: https://www.benefits.va.gov/HOMELOANS/documents/docs/cash_out_refinance_user_guide.pdf

[18] R. Tibshirani, “Regression shrinkage and selection via the lasso,” JRSS-B, 1996. Available: https://webdoc.agsci.colostate.edu/koontz/arec-econ535/papers/Tibshirani%20%28JRSS-B%201996%29.pdf

[19] K. B. Dunn and J. J. McConnell, “Valuation of mortgage-backed securities,” Journal of Finance, vol. 36, no. 3, pp. 599–617, 1981.

[20] E. S. Schwartz and W. N. Torous, “Prepayment and the valuation of mortgage-backed securities,” Journal of Finance, vol. 44, no. 2, pp. 375–392, 1989.